Table Bondhu - Complete Project Development Guide

Project Overview

Table Bondhu is an ESP32-based smart desk companion featuring an anime-style virtual character on a 1.8-inch TFT display. Features include animated emotions, reminders, alarms, ambient light awareness, voice interaction, and cloud connectivity.

Hardware Components

ESP32 microcontroller, 1.8-inch TFT display, MAX9814 microphone, LDR sensor, push button, Wi■Fi/Bluetooth connectivity.

Development Phases

Phase 1: Character engine and display. Phase 2: Character sprite creation. Phase 3: Animation system (blink, breathing, talking). Phase 4: Graphics pipeline. Phase 5: Real-time animation engine. Phase 6: Clock and UI. Phase 7: Reminder system. Phase 8: LDR integration. Phase 9: Voice commands. Phase 10: Wi■Fi/mobile/cloud integration.

Character Design

Use chibi/pixel-art style with large eyes and simple expressions. Emotions: Idle, Happy, Sleepy, Listening, Thinking, Sad, Excited, Surprised.

Recommended Communication Architecture

Version 1: Web dashboard over Wi■Fi. Version 2: BLE mobile control. Version 3: Cloud-connected architecture using Wi■Fi and a mobile application.

Chosen Setup Method: Bluetooth Provisioning

On first boot, ESP32 starts BLE advertising. Mobile app discovers TableBondhu device. User selects Wi■Fi network and enters password. Credentials are sent via BLE and stored in ESP32 NVS. ESP32 connects to Wi■Fi and cloud services.

BLE Provisioning Flow

Power On -> No Wi■Fi Found -> Start BLE Setup -> Phone Connects -> Send SSID/Password -> Save Credentials -> Connect Wi■Fi -> Connect Cloud.

Factory Reset

Hold push button for approximately 10 seconds. Stored credentials are erased. Device re-enters Bluetooth setup mode.

Cloud Backend Suggestions

MQTT (recommended), Firebase, or Supabase. MQTT is ideal for reminders, commands, and real-time status updates.

Recommended Development Order

1. TFT Display 2. Character Animation 3. BLE Provisioning 4. Wi■Fi Connectivity 5. Cloud Connectivity 6. Reminder System 7. Mobile Application 8. Voice Features